

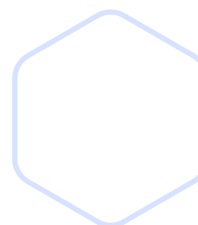
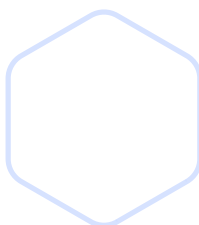
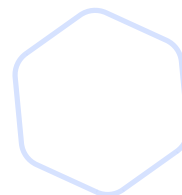
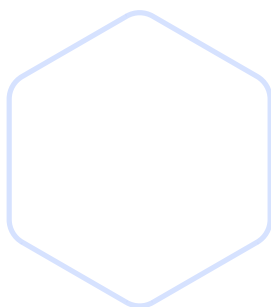
IMPORTANT

Concepts For Interview

321

\$

967548



<D76C@

?>>

Interview Questions & Explanations



Disclaimer

Everyone learns uniquely.

%

Learn BackEnd in a structured manner and master it by practically applying your skills.

%

%

This Doc will help you with the same.

What are the different languages present in DBMS?

The four types of DBMS languages are as follows:

Data Manipulation Language (DML): It is used to manipulate the data and consists of the command for the same. E.g.: SELECT, INSERT, DELETE, UPDATE, etc.

Data Definition Language (DDL): It is used to define and update the data. E.g.: TRUNCATE, ALTER, DROP, CREATE, RENAME, etc.

Data Control Language (DCL): It is used to control the access to the data. E.g.: GRANT, REVOKE, etc.

Transaction Control Language (TCL): It is used to handle the data transactions. E.g.: COMMIT, ROLLBACK, etc.

What are ACID properties?

ACID properties are a set of properties that ensure reliable and secure transactions among databases. To maintain data consistency, ACID properties are followed. ACID stands for Atomicity, Consistency, Isolation, Durability.

Atomicity: Either the entire transaction takes place at once or not at all.

Consistency: The database must be consistent before and after a transaction

Isolation: No other transaction can alter the data during a transaction is in progress

Durability: The transactions made should be durable and must persist

What is normalization? Explain the different types of normal forms.

Normalization is the technique that reduces data redundancy and eliminates insertion, updation and deletion anomalies. Normalization is the process of dividing the larger table into smaller tables and linking them through relationships. It is the process of organizing data in a database.

- **Insertion Anomaly:** Insertion Anomaly is when one cannot insert a new tuple into a relationship due to lack of data.
- **Deletion Anomaly:** Delete anomaly is where the deletion of data results in the unintended loss of some other important data.
- **Updation Anomaly:** Updation anomaly is when an update of a single data value requires multiple rows of data to be updated.

Different types of normal forms:

- **1NF:** It is known as the first normal form. A relation is said to be in 1NF if it contains an atomic value.
- **2NF:** It is known as the second normal form. A relation is said to be in 2NF if it is in 1NF and each non-prime attribute is fully functionally dependent on the primary key.
- **3NF:** It is known as the third normal form. A relation is said to be in 3NF if it is in 2NF and there is no transitive dependency.

- **BCNF:** It is known as Boyce Codd Normal Form which is a strict version of 3NF. A relation is said to be in BCNF if it is in 3NF and for every functional dependency $X \rightarrow Y$, X is a super key of the table. It is also called the 3.5 Normal Form.
- **4NF:** It is known as the fourth normal form. A relation is said to be in 4NF if it is in BCNF and there is no multivalued dependency in the table.
- **5NF:** It is known as the fifth normal form. A relation is said to be in 5NF if it is in 4NF and it cannot be further decomposed into smaller tables.

What is an ER diagram?

ER diagrams stand for **Entity Relationship Diagram**. It is a diagram that displays the different entities and the relationship among them stored inside the database.

ER diagram provides a logical structure of the database. Creating an ER diagram for the database is a standardized procedure and is done before implementing the database.

ER diagram is based on three concepts:

Entities: It can be defined as an object having some properties. They are represented using rectangles. Eg: Car

Attributes: The properties of an entity are called attributes. They are represented using ellipses. Eg: Car name, car mileage, car type, etc

Relationships: Relationships are how the entities are related to each other. They are represented using lines.

